CSE 4074 Computer Networks

Fall 2022 Final Exam

25.01.2022 Wednesday

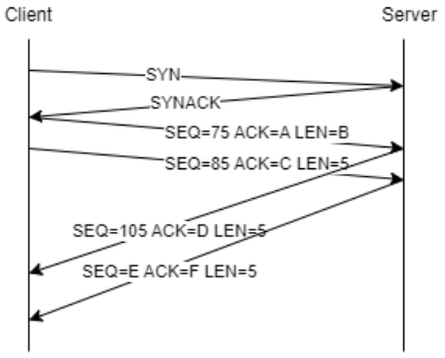
Duration: 40 Minutes for Part 1, Total: 105 Minutes.

Q1	Q2	Q3	Q4	Q5	Q6	SUM
/45	/10	/15	/24	/8	/23	/100

Group-A

Q-1. (45 pts) Multiple choice questions.

Consider the following TCP flow between a client and a server. Answer questions 1-3.



1) What are the values of A, B and C?

a. 105, 10, 105
b. 105, 10, 110
c. 85, 10, 90
d. 85, 5, 90
e. 1, 5, 1

Not b, because client didn't receive any segment between SEQ=75 & 85, so expected SEQ # = ACK# doesn't change

2) What are the values of D, E and F?

a. 75, 110, 85
b. 85, 110, 90
c. 75, 115, 85
d. 85, 110, 85
e. 1, 110, 1

3) Which of the following is **not true**?

a. Initial SEQ number of the client is learned by the server via the SYN message.
b. If the segment with SEQ=105 is not received successfully due to bit errors, the client should retransmit the segment with SEQ=75. No need due to cumulative acknowledgment.
c. If the segment with SEQ=75 is not received successfully due to bit errors, the first segment from the server to the client would have an ACK number of 75.
d. Third step of the 3-way handshaking contains client to server data.
e. In the SYNACK segment, both the SYN bit and ACK bit is set to 1.

4) Consider that a Go-Back-N sender sends four packets. The second packet is lost. Other packets are received by the receiver. The receiver sends three ACKs and only the last one is received by the sender. Others are lost. There is no timeout event. Which packets will be retransmitted by the sender?

a. Only the second packet.
b. The second and the third packets.
c. The first, the second and the third packets.
d. The second, the third and the fourth packets.
e. All the four packets.

No need to send the first one. Because the last ACK will indicate that it is received, due to cumulative acknowledgement.

5) Suppose that EstimatedRTT for a TCP sender is initially 100ms, and exponential weighted moving average is used (with alpha=0.125) to update this value. The next sample RTTs are A, B and C milliseconds respectively. If the final value of the EstimatedRTT is less than 100 ms, which of the following are possible values of X, Y and Z?

a. A=100, B=120, C=80
b. A=90, B=90, C=120
c. A=90, B=100, C=110
d. A=80, B=100, C=119
e. A=100, B=100, C=100.

80 is newer, hence it has more effect. The exp. moving avg will be lower than 100, although the mean is 100.

6) Which of the following is a **false relation** between a TCP header field and the service it is used for?

a. Cwnd : Congestion Control
b. Rwnd : Flow Control
c. 32-bit Sequence number: Reliability
d. FIN: Connection Management
e. 32-bit ACK number: Error Detection.

ACK is used for Reliability. For error detection, we use Checksum.

7) Which of the following is **true** for TCP Slow Start? (MSS= Maximum Segment Size)

a. Congestion window is incremented by 1 MSS in every RTT.
b. Congestion window is incremented by 1 MSS for every ACK received.
c. Congestion window is incremented by MSS/RTT for every ACK received.
d. Congestion window is incremented linearly.
e. Congestion window is not incremented until TCP Slow Start ends.

This way it doubles per RTT.

8) Which of the following is **not true** for TCP CUBIC? (W_{max} = Sending rate at which packet loss is detected)

a. It is the default TCP version in Linux.
b. It is the most popular TCP version for popular Web Servers.
c. Typically it provides higher throughput compared to classic TCP.
d. After multiplicative decrease, congestion window initially ramp to W_{max} faster, but then approach more slowly.
e. After multiplicative decrease, congestion window is incremented cubically until it reaches W_{max}.

9) Which of the following is **false**?

a. In network-assisted congestion control, routers indicate congestion by setting ECN bits in IP header.
b. In network-assisted congestion control, destination sets ECE bit in TCP header in case of congestion.
c. When TCP and UDP flows share the same bottleneck link, there could be a fairness problem.
d. When only TCP sessions share the same bottleneck link, Additive Increase Multiplicative Decrease mechanism always guarantees fairness among network applications.
e. There are some TCP congestion control mechanisms that reduce sending rate before detecting packet loss.

It doesn't guarantee always. For example, when apps use different number of connections, when RTTs are different, etc.

10) Which protocol is used to learn the address of the local DNS server?

a. ICMP
b. DHCP
c. SNMP
d. NETCONF
e. DNSSEC

11) Suppose that IP address of host A is 192.167.3.136 and subnet mask is 22. Which of the following is a possible IP address of host B that is in the same subnet?

I. 192.167.1.235
II. 192.167.3.046
III. 192.167.3.235
IV. 192.167.7.168

a. Only II
b. Only III
c. II and III
d. I, II and III
e. II, III and IV

Since subnet mask is 22, last 10 bits should be different, but first 22 bits should be the same. They are the same for the first three.

12) Which of the following is **not true** about IPv6?

a. It is carried inside IPv4 packets if the routers do not support IPv6.
b. It provides fragmentation when packets are too big.
c. It does not provide any field for error detection.
d. The header size is fixed, and it is 40 bytes.
e. 80% of the header consists of the source and destination addresses.

fragmentation is not provided by IPv6.

13) In a router, where can queuing delay and loss occur?

I. Input port
II. Output port
III. Switching fabric

a. Only I
b. Only II
c. I and II
d. I and III
e. I, II and III

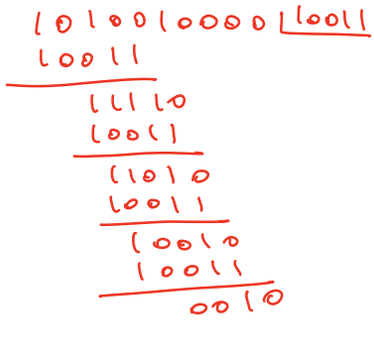
There are buffers in the input port and the output port.

14) Which of the following is **not true** about generalized forwarding?

a. It is possible to configure a router as a firewall.
b. It is possible to configure a router as a link-layer switch.
c. Drop, copy, modify and log packet actions are possible.
d. It is possible to forward TCP segments and UDP segments with the same source-destination pair to different output ports.
e. It is not possible to forward HTTP packets and SMTP packets with the same source-destination pair to different output ports.

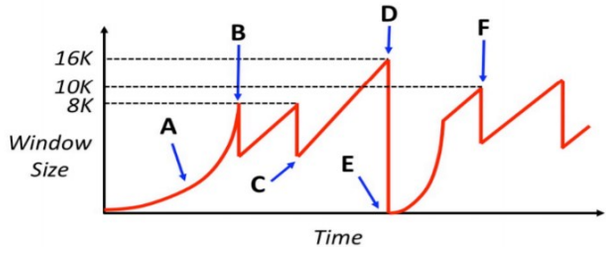
15) If data is 101001 and the generator is 10011 what would be the CRC bits?

a. 010
b. 0010
c. 001
d. 0011
e. 011



Since # bits of CRC = # bits of Generator minus 1, it should be 4 bits.

Q-2 (10 pts) Consider the following graph, where y axis is the TCP congestion window size (in bytes) of the sender (note that the graph is not drawn to scale). TCP Reno is used.



(a – 4 pts) The window size of the TCP sender decreases at several points, including those marked by B and D. What happens at B? What happens at D?

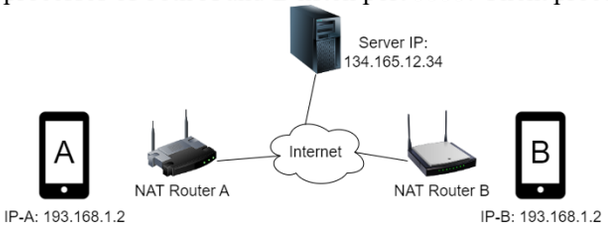
B → 3 duplicate ACK D → Timeout.

(b – 6 pts) Consider the round trip time (RTT) between the sender and receiver is 100 ms and the maximum segment size (MSS) is 1K bytes (so 8K window size = 8 MSS window size). Ignore any other delay other than RTT.

i. How much time has progressed between C and D? From 4K to 16K, 12RTT, i.e. 1200ms.

ii. How much time has progressed between E and F? 1K→2K→4K→8K→9K→10K 5RTT→500ms.

Q-3. (15 pts) Consider the following scenario where A and B wants to communicate via a peer-to-peer application. Both of them have the same local IP address (193.168.1.2). Both NAT Router A and NAT Router B have the same local IP address of 193.168.1.1. Their WAN-side IP addresses are 210.202.12.15 and 183.175.63.4 respectively. Typically P2P applications include both server processes (to be contacted) and client processes (to contact). Server processes of both A and B listen port 5555. Client processes of A and B use port 1234 and 4321 respectively.



(a – 4 pts) Describe how A can establish a P2P connection with B without statically configuring the NAT tables, with the aid of Server.

A establishes a connection with the server.
B establishes a connection with the server.
Server acts as a relay.

(b – 6 pts) After establishing P2P connection, what would be stored in NAT tables of NAT Router A and NAT Router B?

NAT Table of A

LAN side	WAN side
193.168.1.2, 1234	210.202.12.15, newport1

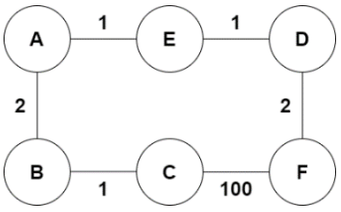
NAT Table of B

LAN side	WAN side
193.168.1.2, 4321	183.175.63.4, newport2

(c – 5 pts) Can you configure NAT via Software Defined Networking? If no, why? If yes, how would you set match+action in OpenFlow?

Yes. For outgoing packets, src IP, port# is modified with WAN IP and new port#. For incoming packets, dest IP, port # is modified with LAN side IP and port #.

Q-4. (24 pts) Consider the following graph:



(a – 8 pts) Show the operation of Dijkstra's (Link State) algorithm for computing the least cost path from **B** to all destinations. How does B set its forwarding table entries for all destinations if the traditional destination-based forwarding is used?

	A	B	C	D	E	F
B	B,2	0	B,1	∞	∞	∞
BC	B,2			∞	∞	C,101
BCA				∞	A,3	C,101
BCAE				E,4		C,101
BCAED						D,6

Forwarding Table:

A	(D, A)
C	(B, C)
E	(B, A)
D	(B, A)
F	(B, A)

(b – 12 pts) Suppose that Distance Vector Routing is used and all the distance vectors are converged. Show the distances vectors of A and E. Describe how removal of AE link causes count-to-infinity problem. Which one sends more DV messages, A or E? Why?

$DV_A = <$ A: 0, B: 2, C: 3, D: 2, E: ~~1~~ 5 ~~9~~ 13 .. 106, F: 4 $>$

$DV_B = <$ A: 2, B: 0, C: 1, D: 4, E: ~~3~~ 7 ~~11~~ 15 .. 108, F: 6 $>$

$DV_E = <$ A: ~~1~~ 3 ~~5~~ ~~7~~ ~~9~~ .. 106, B: 3, C: 4, D: 1, E: 0, F: 3 $>$

$DV_D = <$ A: ~~2~~ 4 ~~6~~ ~~8~~ ~~10~~ .. 107, B: 5, C: 0, D: 1, E: 2, F: 2 $>$

A increments 2 by 2, while E increments 1 by 1. So E sends more.

(c – 4 pts) In BGP, you may not always use the shortest (minimum-cost) paths. Briefly describe two reasons.

(Hint: Second reason is related to hot-potato routing)

- Due to policy restrictions, you may not learn the shortest path.
- An AS may decide to choose a path with min. intra-AS cost, although it takes a longer path outside the AS (hot-potato routing)

Q-5. (8 pts) Please answer the following questions related to multiple access protocols.

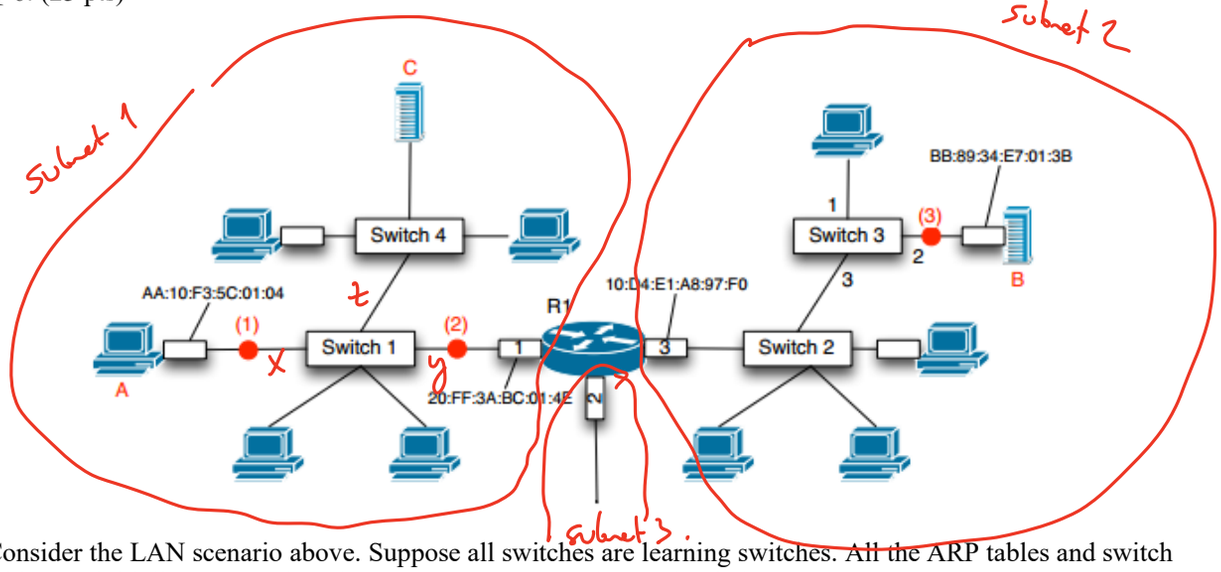
Please answer the following questions:

(a – 4 pts) In CSMA/CD, is there any relation between propagation delay and collision probability? (give also the reason.) If propagation delay increases, collision probability increases. Because collisions will be detected lately.

(b – 4 pts) For wide area cellular networks (such as 3G or LTE), channel partitioning MAC protocols are more appropriate than random access MAC protocols. What could be a reason?

In wide area cellular networks, there are many users and high load. Random Access MAC protocols could result in too many collisions. Channel partitioning MAC protocols will be collision-free. (This is one reason).

Q-6. (23 pts)



Consider the LAN scenario above. Suppose all switches are learning switches. All the ARP tables and switch tables are initially empty. There is no NAT. Answer each question below.

(a – 4 pts) Considering all the hosts/interfaces shown in the figure, at least how many different IP addresses are needed? How many different subnets these IP addresses belong to?

14 interfaces, so 14 IP addresses are needed. There are 3 subnets shown in the figure.

(a – 3 pts) Suppose that A wants to send a packet to B. Consider an ARP request sent by node A. Whose IP-to-MAC address translation is being queried?

R1 - interface 1.

(b – 4 pts) Continue from question (a). What would be the destination MAC address of the ARP request frame sent by A? Which node(s) will receive this ARP request?

Dest. MAC: FF.FF.FF.FF.FF.FF. All the nodes in the left subnet. (subnet 1)

(c – 4 pts) Continue from question (b). After A receives ARP reply, what is contained in switch tables of switch 1 and switch 4?

switch 1: MAC of A, interface x
MAC of R1-1, interface y

switch 4: MAC of A, interface z

x, y, z shown in the figure.

f) (4 pts) Let's say that a packet passes over points (1), (2) and (3). Is there any change in the IP address fields of this packet? What about MAC address fields of the frame(s) containing this packet? Describe the changes.

Since there is no NAT, IP src & IP dest do not change.
In (1) and (2) src MAC: MAC of A, dest MAC: MAC of R1-1.
In (3), src MAC: R1-3 dest MAC: MAC of B.

f) (4 pts) Consider the datagram being sent from A to C; neither A nor C have sent any frames or datagrams in the network before. (Ignore any ARP requests sent before the frames.) How many of hosts in the network receive the frame containing the datagram sent by A? What about the reply sent from C to A? Explain your answer briefly.

While A is sending a frame, since switch tables are empty, both switch 1 and switch 4 flood. So all 6 interfaces in subnet 1 (except A) will receive the frame.

When C replies to A, since switch 4 and switch 1 learned the interface for A, they will directly send the frame, and only A receives.